

Main figure legends

Figure 1. PHLDA3 is over-expressed in thyroid carcinoma and validated externally. (a) Whole-transcriptome volcano plot of TCGA-THCA tumour-versus-normal differential expression; PHLDA3 highlighted. (b) Pan-cancer tumour-versus-normal PHLDA3 (TCGA tumour vs TCGA + GTEx normal); THCA highlighted (fourth-largest up-regulation). (c) External validation in GSE33630 (normal thyroid, PTC, ATC). The tumour-versus-normal ROC (AUC 0.965) is shown but interpreted as confirmation of over-expression rather than a diagnostic test (Methods 2.2).

Figure 2. PHLDA3's lymph-node-metastasis association is shared with the kinase-driver landscape. (a) Base multivariable logistic-regression forest for N1 (PHLDA3 + age + sex + T stage). (b) PHLDA3 odds ratio for N1 across nested models (base $\rightarrow$ + BRAF $\rightarrow$ + BRAF + RET/NTRK/ALK fusion; all fitted on the identical 432 patients), attenuating to non-significance once fusion status is added. (c) PHLDA3 expression across four mutually exclusive driver classes: comparably high in BRAF^{V600E} and kinase-fusion-positive tumours, low in RAS-mutant and driver-negative tumours. (d) Incremental 10-fold cross-validated AUC—PHLDA3 adds Δ AUC $\approx$ 0.001 over a driver-aware model (age + sex + T + BRAF + fusion).

Figure 3. PHLDA3 marks the kinase-driver-defined landscape of thyroid carcinoma. Across four mutually exclusive driver classes (BRAF^{V600E}, RET/NTRK/ALK fusion-positive, RAS-mutant, WT/other): (a) PHLDA3 expression; (b) nodal-metastasis (N1) frequency, highest in the BRAF^{V600E} and fusion-positive classes; (c) PHLDA3-promoter methylation (cg04055835), lowest in those same two classes; (d) integrated PHLDA3 expression versus promoter methylation coloured by driver class (Spearman $r = -0.62$), with the BRAF/fusion classes clustering at high expression/low methylation.

Figure 4. PHLDA3 and the immune microenvironment, adjusted for tumour purity. Correlation of PHLDA3 with ssGSEA immune-cell infiltration (Danaher signatures) and immune-checkpoint genes, shown as raw and tumour-purity-adjusted (ESTIMATE partial Spearman) coefficients; associations surviving purity adjustment are highlighted.

Figure 5. PHLDA3 is induced in malignant thyroid cells and tracks p53-programme activity. (a) Percentage of cells expressing PHLDA3 by cell type in thyroid tumours (GSE193581), malignant cells highlighted. (b) PHLDA3 in malignant (PTC/ATC) versus normal epithelial cells (cell-level, descriptive). (c) PHLDA3 versus a p53-target module score within malignant cells: cell-level scatter (grey, descriptive) overlaid with the sample-level pseudobulk correlation (one point per malignant sample; Spearman $r = 0.58$, $P = 0.030$, $n = 14$). (d) PHLDA3–p53 coupling across cell types. (e,f) Sample-level comparison (each point = one sample) of the percentage of cells detected (e) and mean expression (f), addressing single-cell pseudoreplication.

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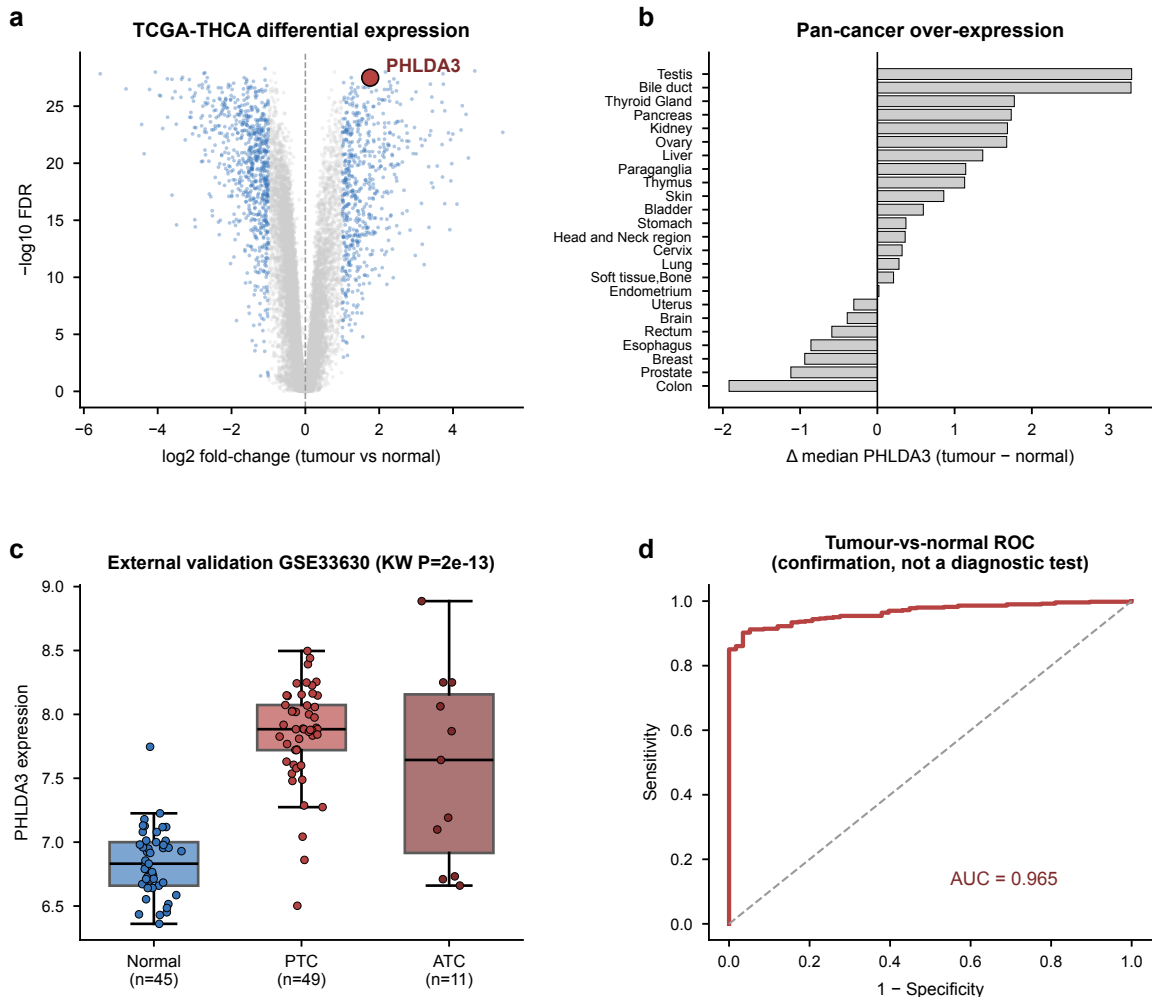


Figure 2. PHLDA3's lymph-node-metastasis association is shared with the kinase-driver landscape

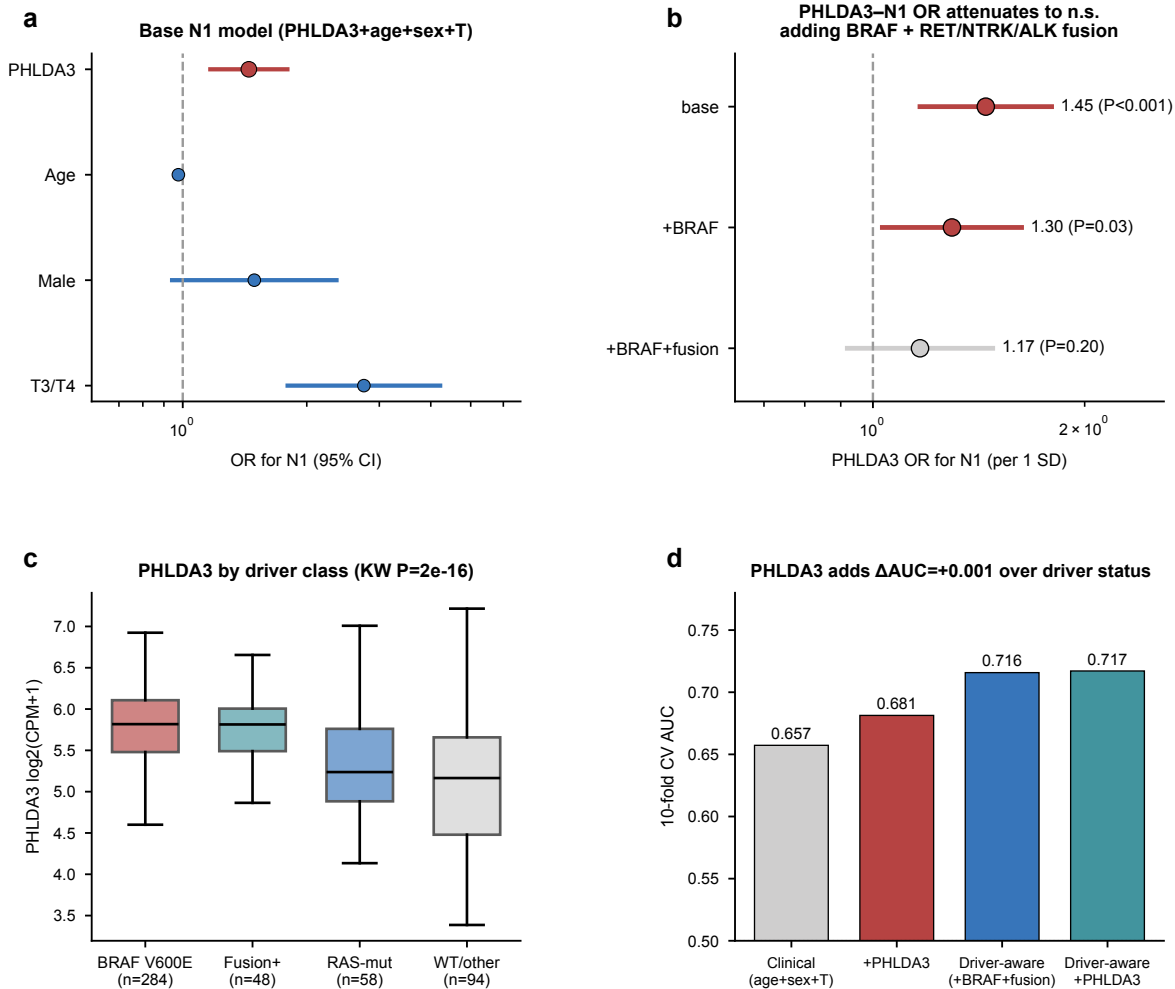


Figure 3. PHLDA3 marks the kinase-driver-defined landscape of thyroid carcinoma

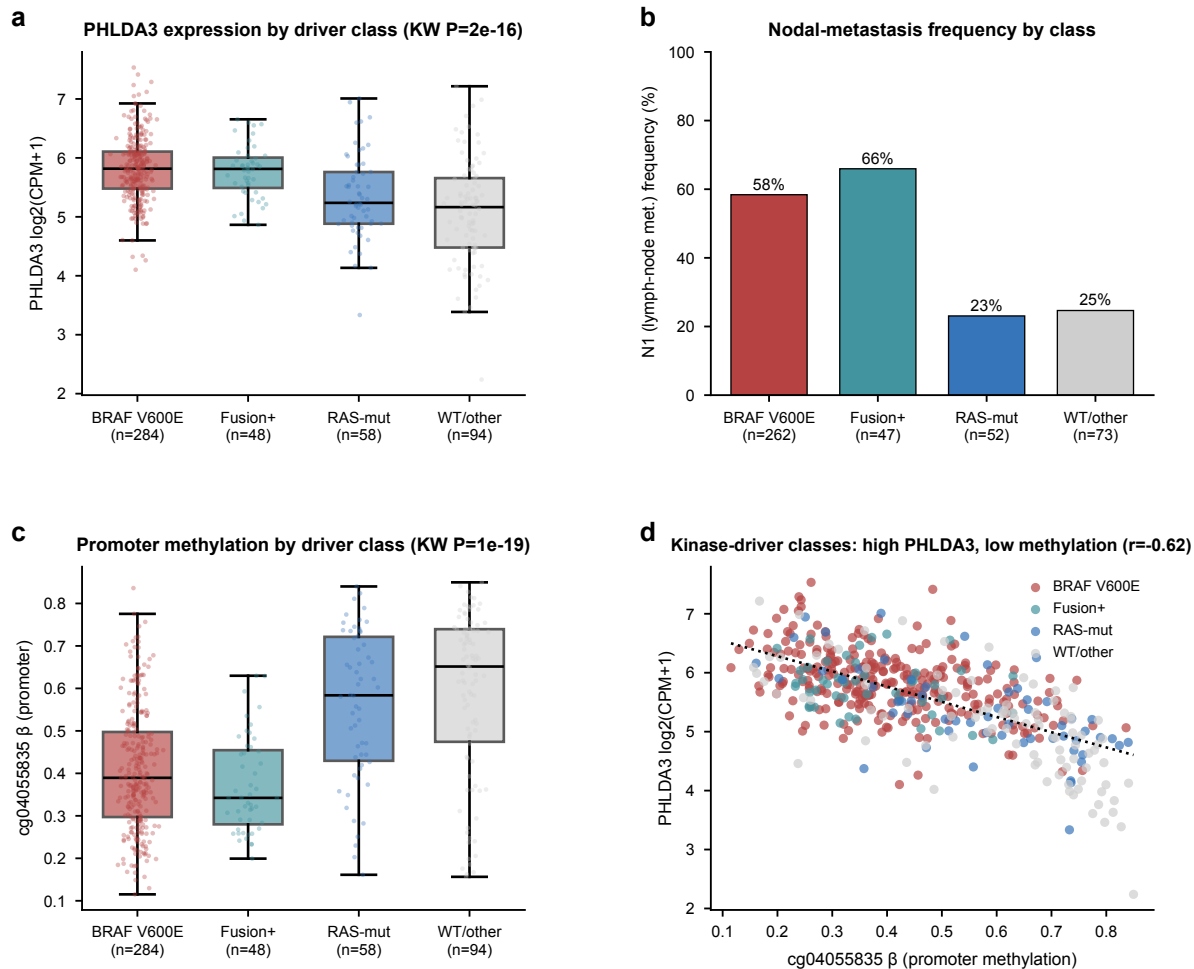


Figure 4. PHLDA3 and the immune microenvironment, adjusted for tumour purity

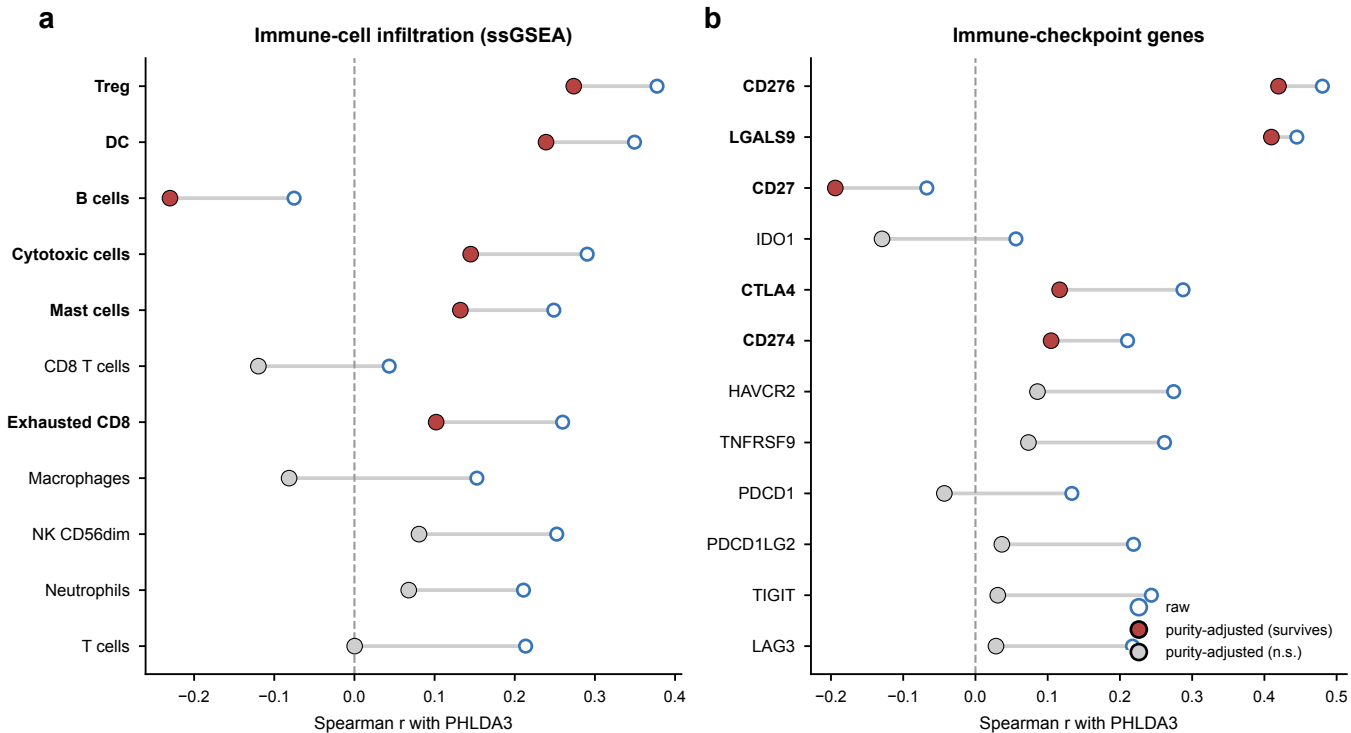


Figure 5. PHLDA3 is induced in malignant thyroid cells and tracks p53-programme activity

